Forecasting HDB Rental Rates in Singapore

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***Abstract*—We predict monthly rents for 50,000 Housing and Development Board (HDB) rental approvals from March 2025 to July 2026 using 150,000 approvals from January 2021 to March 2025. Prediction requires temporal extrapolation since 99.2% of test records occur after the last labeled month. In a chronological baseline holdout, the month counter has zero permutation importance because tree splits on this feature cannot extrapolate beyond the training range. We estimate a multiplicative market index by iterative ratio updates over rental groups and months, then forecast it with a damped trend. LightGBM and CatBoost learn rent divided by this index. Their rescaled predictions are combined with histogram-based gradient boosting (HistGB) trained on unadjusted rent. Removing the index increases root mean squared error (RMSE) by 86.4 in a matched LightGBM ablation. Across 11 model classes, RMSE varies by S$32.2 under a fixed index setup. LightGBM tuning reduces it by S$3.5. Rows sharing all recorded predictors with another row constitute 27.6% of rolling-validation observations. On this subset, estimated pooled within-group noise variance equals 98.8% of one LightGBM configuration's mean squared error, with a corresponding noise standard deviation of S$509. This conditional estimate does not establish an error floor for other rows or the hidden test set. Our best public-leaderboard RMSE is S$430.76, obtained by the ensemble with index shrinkage 0.7.**

***Index Terms*—rental price prediction, time series decomposition, gradient boosting, temporal validation, irreducible error**

I. Introduction

HDB rents rose steeply after 2021, raising housing costs for households that do not own a flat, including those not yet eligible to buy one and those waiting for a new flat to be completed [1]. A rent prediction lets a prospective tenant compare an asking rent with the expected rent of flats with the same recorded attributes. A landlord can see how the expected rent varies with location, flat size, and age. A housing authority can follow the market level after adjusting for the types of flats being rented, which is needed to judge whether an affordability measure has changed rents.

The task is to predict the monthly rent of 50,000 Housing and Development Board (HDB) rentals approved between March 2025 and July 2026 from 150,000 labeled rentals approved between January 2021 and March 2025. The test set is separated from the training set in time rather than sampled at random, and 99.2% of its rows fall after the last labeled month. Over the labeled period the mean monthly rent rose from S$2,049 in January 2021 to S$3,163 in March 2025, an increase of 54%. A model must therefore predict the market level up to 16 months beyond the training data as well as the differences between flats within a month.

The temporal and cross-sectional components require different treatment. A standard regression tree partitions the training feature range into regions with constant predictions. With other predictors held fixed, values beyond the largest observed month follow the same branches, so adding a month feature cannot extrapolate a continuing trend. Other inputs, including time-dependent ages and macroeconomic variables, can still change predictions. Cross-sectional accuracy is limited by the recorded attributes. Rentals of the same flat type in the same block and month have identical predictors but different rents. Section VIII estimates the variation within these groups.

We address five research questions, namely (1) how to forecast the market level over 16 months, (2) whether location features derived from past rents improve accuracy beyond coordinates, (3) which feature groups and model classes contribute predictive information, (4) how validation should represent the forecast horizon, and (5) how much residual variation remains among records with identical predictors. Figure 1 summarizes the method. Rent is modeled as a flat-specific base multiplied by a monthly market index. The index captures changes in the common rent level before regression on flat attributes. Removing it raises LightGBM validation RMSE by 86.4. The five best of eleven model classes differ by 5.9 under a fixed comparison setup.

II. Data and Exploratory Analysis

A. Task and Data Sources

The task is a Kaggle competition [1] based on HDB rental approval records published on data.gov.sg [2]. Each row records the approval month, town, block, street, flat type, flat model, floor area, lease commencement year, furnishing status, and upfront fee. Training rows also record the monthly rent in Singapore dollars. Submissions are evaluated by root mean squared error (RMSE). Rent-error metrics are reported in Singapore dollars unless expressed as ratios.

The competition supplies six auxiliary tables. They contain block characteristics and coordinates for 9,464 blocks and the locations of 243 Mass Rapid Transit (MRT) station records, 113 shopping malls, and 337 schools. They also contain monthly Certificate of Entitlement (COE) premiums for four vehicle categories and daily share prices of 77 Singapore-listed companies. Among the auxiliary sources, only the COE and share-price series vary with approval month and extend into the test period.

We added the HDB Property Information register [3], which covers 13,357 blocks and records dwelling counts by flat type, rental-unit counts, and block facilities. It contains no transaction rents, so the join does not supply target values. As a static register, it may not represent each block's historical

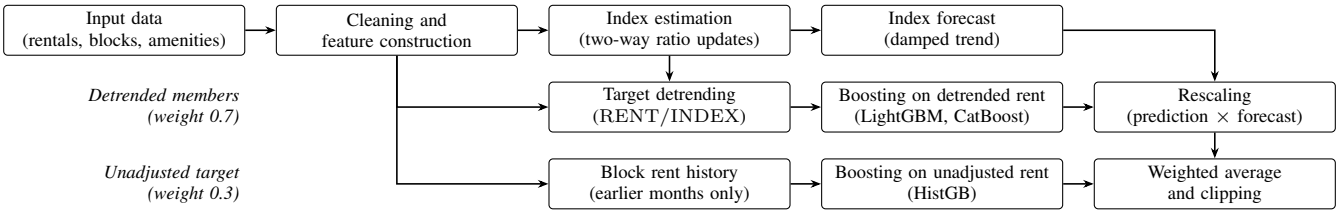


Figure 1. The pipeline combines two index-adjusted models with a model trained on unadjusted rent. The top row constructs features, estimates monthly market indices by flat type, and forecasts them over the prediction window. LightGBM and CatBoost learn rent divided by the fitted index, and their predictions are multiplied by the forecast index. HistGB learns unadjusted rent using earlier block–flat-type rents as an additional feature. The ensemble assigns weight 0.7 to the mean index-adjusted prediction and 0.3 to HistGB. All components are refitted using each outer fold's training data.

condition. The public rental approval register was excluded because its coverage extends to July 2026 and includes test-period rents.

B. Cleaning and Target Distribution

Neither the training file nor the test file contains missing values. Two columns are constant. Column `FEE` equals zero and column `FURNISHED` equals "yes" in all 200,000 training and test rows. A constant column cannot distinguish one rental from another, so both are dropped. Column `FLAT_TYPE` has eleven distinct values in the training file because each numbered type is written in two formats, for example "4 room" and "4-room", whereas the test file uses only the hyphenated format. After every value is mapped to the hyphenated format, six flat types remain. The 1-room type is rare, with 42 training rows.

Rent ranges from S$300 to S$7,600, with a mean of S$2,719 and a standard deviation of S$750. The median rent increases from S$1,450 for 1-room flats to S$3,000 for 5-room and executive flats, but floor area alone explains only about 11% of the variance. The annual mean rent was S$2,121 in 2021, S$3,036 in 2023, and S$3,110 in 2024, so most of the increase occurred before 2024.

We evaluated outlier removal and logarithmic transformation. A Tukey fence, defined by extending 1.5 interquartile ranges below the first and above the third quartile, flags 752 rents (0.5%) outside S$450–S$4,850. One third have other records with the same flat type, block, and month. The median absolute deviation from their group mean, which includes the flagged rent, is S$333. This comparison does not establish that the records are erroneous, so they were retained. Rent has skewness $+0.39$, whereas log rent has skewness -0.42. The log-target configuration increases RMSE by 5.4 and triples the magnitude of mean error, so subsequent models use rent in levels. Figure 3 shows both distributions.

Joining the block table on town, block, and street initially matched only 79.5% of training rows, and all mismatches had the same cause. The block table stores street names in lower case, whereas 20.5% of rows in the rental files use title case, for example "Ang Mo Kio Avenue 3" and "ang mo kio avenue 3". Converting street names to lower case raises the match rate to 100%.

The inconsistent case also affected every quantity grouped by street and block, because the two spellings were treated as different blocks. The training file contains 17,413 distinct combinations of town, block, and street but only 9,406 physical blocks, and 8,007 of these blocks appear under both spellings. Without normalization, the rental history of most blocks would be split in two, which would distort both the block history feature of Section VI-B and the index units of Section V.

C. Variation by Month and Location

We first assessed variation associated with approval month and location. Replacing each rent by its approval-month mean explains 32.1% of sample variance. Using town–flat-type means explains 21.9%. Combining the groupings explains 57.6%. These descriptive, in-sample calculations indicate complementary temporal and cross-sectional information. They motivate separate modeling of the market level and flat attributes, but do not establish independence or constant relative prices.

Variation also remains among records with identical predictors. In this dataset, floor area, flat model, and lease commencement year are constant within each block–flat-type group. Rentals with the same block, flat type, and month therefore agree on every competition predictor. A deterministic model using these predictors must assign them the same prediction, making the group mean the minimum-squared-error fit to the observed group. A repeated feature group contains at least two records with identical predictors. Such groups contain 26.8% of training rows. Their observed within-group RMSE is S$343. Correcting for estimation of one mean per group gives a pooled noise standard deviation of S$464, approximately 17% of mean rent. The estimator and assumptions are given in Appendix H. This variation can reflect unobserved attributes or other noise [5]. For example, seven 3-room, 65-square-metre rentals in block 4 Holland Close approved in January 2025 have rents of S$3,300, 3,200, 3,050, 3,000, 2,800, 2,200, and 1,000.

III. Validation Protocol

The main model comparisons use a common chronological protocol with four requirements.

Training months precede validation months, reflecting prediction of future approvals rather than interpolation within the observed period [6], [7]. Splits occur at month boundaries so that records from the same block and month cannot appear

on both sides. Each rolling validation window covers the 16-month future forecast horizon. Quantities derived from rent, including the index, block history, and target encodings, are recomputed from each fold's training rows. This prevents validation rents from entering fitted features or index forecasts.

We use rolling-origin evaluation [6]. The training period expands to March, July, and November 2023, and each fitted model predicts the following 16 months (Table 4). Each validation fold contains approximately 47,000 rows. We report mean fold RMSE and mean prediction error, defined as prediction minus observed rent, to distinguish overall error from systematic over- or under-prediction. A fourth origin after March 2024 provides 12 labeled months. Candidates were retained only when they improved both the rolling mean and this confirmation cut. The confirmation cut therefore contributes to model selection and is not an independent test.

The validation windows overlap, and the confirmation window lies within the third rolling window. Their scores are consequently dependent. Difficulty also varies by period. One configuration scores approximately 555, 519, and 483 across the rolling folds. Models are compared on identical observations within each fold. Scores obtained under different windows are not directly comparable.

IV. Baselines and the Extrapolation Problem

We first established two baselines. The first is ordinary least squares (OLS) regression on floor area, lease age in years, town, and flat type. Town and flat type use one binary indicator for each category. Each indicator equals one when a record belongs to that category and zero otherwise. This representation is called one-hot encoding.

A linear model with additive terms assigns each town a single premium that is added to the rent of every flat type, so a town that adds S$300 to a 3-room flat must also add S$300 to an executive flat. It cannot represent an interaction effect, in which the town premium depends on flat size. Hedonic price theory models a dwelling's price as a function of its structural and location attributes without requiring additivity [8]. We therefore also fit gradient-boosted regression trees [9] with the histogram-based implementation in scikit-learn [13] (HistGB), which learns interactions through successive splits, handles categorical columns natively, and assigns missing values to a learned branch.

The boosting baseline uses 12 features. These comprise the four OLS attributes, flat model, three amenity distances, two economic series, a sequential month counter, and calendar month. Distances from each block to the nearest MRT station, mall, and primary school are calculated from coordinates using the haversine formula [14]. Economic features are monthly mean COE premiums and share prices. Street and block are excluded because their high cardinality and median of 13 training rows per block create a risk of overfitting.

OLS uses months after the 85th percentile of approval dates. HistGB uses the latest 15% of date-sorted rows. Their RMSEs are 700.33 and 464.79, respectively. These initial holdouts differ and are shorter than the forecast horizon, so their scores are excluded from the main rolling-fold comparisons.

Permutation importance [15] measures the decrease in the coefficient of determination (R^2) after shuffling one feature on held-out data. Town and flat type have importances of 0.352 and 0.261, followed by floor area at 0.134 and nearest-MRT distance at 0.012 (Table 6). Correlated predictors complicate interpretation of individual importance scores [16]. These values measure reliance of the fitted model rather than causal contributions to rent.

The month counter has zero importance. The two economic series have marginally negative importance. Each learned threshold on the month counter lies within its training range, and every held-out value is at or beyond that range. Permuting these values therefore leaves the corresponding branches unchanged. This result concerns splits on the month feature. Predictions may still vary with other inputs. It motivates an explicit forecast of the market level rather than reliance on the month counter alone.

V. Multiplicative Market Index

A. Decomposition of Rent

For rental record i approved in month t, we use the approximation

$$\mathrm{RENT}_{it} \approx \mathrm{BASE}_i \, \mathrm{INDEX}_t. \tag{1}$$

The dimensionless index represents the common monthly rent level. It is estimated on labeled months and forecast over the prediction window. LightGBM and CatBoost learn rent divided by the fitted index. Multiplying their predictions by the forecast index returns them to dollars. This transformation reduces changes in the target's market level without establishing stationarity of its full distribution.

Increasing dispersion motivates proportional rescaling. The estimated within-group noise standard deviation rises from S$279 in 2021 to S$588 in 2023. We also evaluated an additive alternative that subtracts a fitted dollar-valued month effect before regression and adds its forecast back. The additive month effect is estimated by alternating group and month mean-residual updates. Both decompositions use one global monthly component, identical structural and block-composition features, fixed LightGBM settings, and the same slope window and damping parameter. Target-based features are excluded from both. Rolling RMSE is 519.41 for the multiplicative decomposition and 525.56 for the additive decomposition. Confirmation RMSEs are 467.04 and 470.68, respectively. These settings were selected for the multiplicative pipeline, so the additive alternative was not independently tuned.

The estimator adjusts for observed dwelling characteristics, as in hedonic price analysis [8], through groups defined by street, block, and flat type. Starting with an index of one, twelve iterations alternate between each group's mean rent divided by the current monthly index and each month's mean rent divided by the current group bases. The index is normalized after each iteration so that the last training month equals one. These arithmetic ratio updates operate on

rent in levels. Adjustment for changing rental composition assumes approximately stable group bases. It does not remove unobserved within-group composition changes. Alternative grouping definitions change RMSE by at most 0.75.

In matched ablations, training on unadjusted rent increases rolling RMSE by 86.4 for LightGBM and 140.0 for CatBoost. Mean errors are S$−207 and S$−333, respectively. These results support explicit adjustment for the market level under the evaluated forecast windows.

B. Forecasting the Index

The index in Equation (1) must be forecast up to 16 months beyond the final training month. A random-walk forecast holds its last value. A linear forecast extends the least-squares slope b fitted over the last w months. The damped forecast [18] is $\widehat{I}_{T+h} = I_T + b\sum_{j=1}^{h}\phi^j$, where T is the last training month, h is the forecast horizon, and $0 < \phi < 1$ reduces successive increments. We crossed w and ϕ on the rolling folds to separate their effects.

The damping parameter has the largest effect. With the slope fitted on twelve months, ϕ values of 0.6, 0.75, 0.85, and 0.95 give RMSEs of 519.87, 521.41, 530.04, and 567.73, while the mean error increases from -4 to $+190$. Weak damping projects the earlier rise into a period in which rents had stopped rising. An undamped linear trend over 18 months gives 672.55, the worst index-forecast configuration tested. The slope window has a smaller effect, since the best setting of each window lies within 0.6 of the overall best. We therefore fit the slope on twelve months with ϕ of 0.6.

On the full training index, the selected forecast increases by 0.3% over 16 months, compared with 2.7% for the undamped 18-month trend (Figure 2). In the tuned window–damping comparison, the selected forecast reduces RMSE from 523.68 for a random walk to 519.87. By contrast, the 86.4-point ablation compares index adjustment with training LightGBM directly on rent. The distinction separates the effect of detrending from the choice of index forecast.

This conclusion depends on the validation period. All three validation windows begin after the 2022 rise, so each fold favours a forecast that expects rents to flatten, and a renewed steep rise would penalize a ϕ of 0.6. Across the forecasts considered, from a random walk to an undamped linear trend, the mean index over the test rows varies by 1.5%, or S$40 of rent (Table 9). This range does not bound the effect on the public score, because the level shift grows with the horizon and its relation to the errors on unlabeled rows is unknown.

A separate submitted pipeline used tuned ridge regression, an L2-regularized linear model, on 71 features and unadjusted rent, without an index. Its predictions correlate with ours at 0.973 within test months, but its mean predicted level rises by 0.80% per month, compared with 0.02%, and ends 15% higher. Its public RMSE is 505.60, versus 430.76 for our alternative-shrinkage ensemble (Section V-C). Both the model and temporal specification differ, so this comparison cannot identify the effect of index forecasting.

C. Segment-Specific Indices

A global index assumes proportional market movements across flat types. Separate fitted indices increase from January 2021 to March 2025 by factors ranging from 1.46 for 2-room flats to 1.61 for executive flats. This difference motivates flat-type-specific indices.

Separate segment indices allow different growth rates but can increase estimation variance in small groups. For example, 2-room flats contribute only 3,098 rentals. We therefore shrink each segment index toward the global index geometrically, giving segment g the index $\mathrm{INDEX}_t(\mathrm{INDEX}_t^g/\mathrm{INDEX}_t)^\lambda$, so that $\lambda = 0$ recovers the global index and $\lambda = 1$ the fully separate one. Segmenting by flat type with $\lambda = 0.5$ gives 519.27 against 519.87 for the global index, and segmenting by region or town gives no improvement.

The final configuration gives RMSEs of 518.94, 518.82, 518.80, and 518.81 for $\lambda = 0.3, 0.5, 0.7, 0.9$, respectively. We selected 0.5 because validation accuracy is insensitive to this range. The 0.7 version scores 430.76 on the public leaderboard, compared with 431.01 for the reported 0.5 version. The observed difference is 0.25 RMSE. Its statistical uncertainty cannot be estimated from the submission files alone because the hidden labels, and hence the paired prediction errors, are unavailable.

VI. Feature Engineering

A. Amenities and Block Composition

After the join repair, we extended the first feature set into eleven groups, covering distances and counts within a radius for MRT stations, malls, and both school levels, coordinates and distance to the city centre, block height and age, remaining lease, floor area with room count, the finer administrative categories, and the calendar month. Each group was removed in turn from the tuned model to measure its contribution.

Amenity features use the supplied locations rather than a dated operational transport network. The MRT table has 243 rows at 192 locations, including repeated interchange coordinates and a record marked open for the planned MRT station serving Changi Airport Terminal 5 [4]. Duplicate coordinates leave nearest distance unchanged but affect second-nearest distance and radius counts on 21% and 27% of rows, respectively. Deduplicating and removing the Terminal 5 station leaves 191 locations and changes RMSE by 0.02. This sensitivity check does not reconstruct the operating network at each approval date. We retain these location features as geographic proxies, with limited interpretation as access to operating stations.

Removing any single point-of-interest group changes RMSE by at most 0.17 from the 519.87 reference. Removing coordinates improves it by 0.23. Removing all ten point-of-interest columns increases RMSE by 1.58, and removing them together with coordinates increases it by 2.24. These results indicate overlapping predictive information. A one-at-a-time ablation understates their joint contribution. Twenty additional distance and count columns improve RMSE by 0.26. The largest individual removal effects are 5.29 for calendar month and 0.64 for floor area.

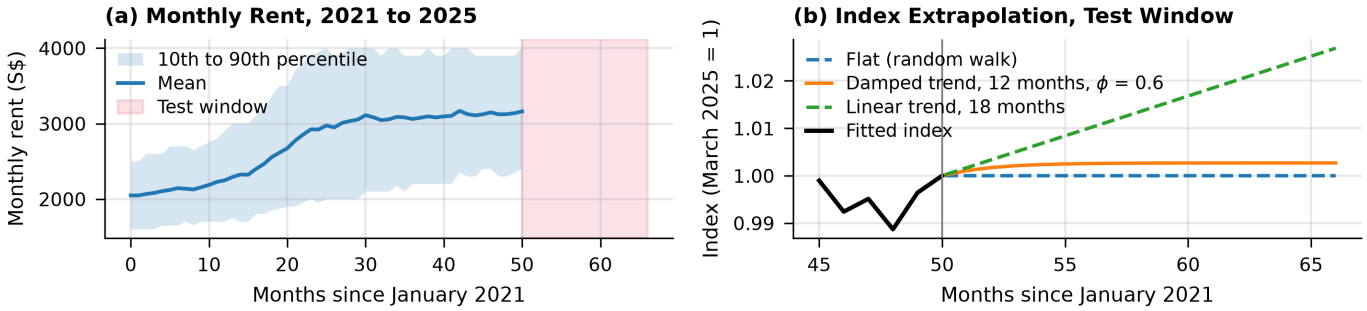


Figure 2. Panel (a) shows monthly mean rent and the band between the 10th and 90th percentiles during the labeled period. The shaded region marks the test period. Panel (b) compares the last five months of the fitted index with three forecasts, normalized to one in March 2025.

Seven features from HDB Property Information describe block composition. They are the dwelling count, the rental, small-unit, and large-unit shares, and indicators for commercial units, a hawker centre, and a multistorey car park. They reduce rolling RMSE from 519.87 to 519.27 and confirmation RMSE from 467.06 to 466.41. They supplement location with information about block composition. The register is unmatched for 159 training rows in 11 blocks and no test rows. These feature values remain missing and are handled by the boosting models.

A horizon feature, the number of months between the last training month and the predicted row, increased the RMSE by 14.3 and was rejected. Within the training data this quantity is confounded with calendar time, so the model used it as a proxy for the date.

We evaluated alternative location and numeric representations. Replacing administrative areas with k-means clusters of block coordinates gives RMSEs of 519.46, 519.47, and 519.24 for $k = 8, 25, 60$, compared with 519.87 for administrative columns. Principal component analysis (PCA), which replaces numeric variables with orthogonal linear combinations ordered by explained variance, increases RMSE by 2.90 with eight components and 0.46 with sixteen. Twenty-four components match the original 25 variables. The largest cluster-based reduction is 0.63 RMSE. PCA offers no reduction in this setup.

B. Block Rent History

We added two features to HistGB trained on unadjusted rent. These are the mean earlier rent for the same street, block, and flat type, and the number of contributing observations. These describe a block–flat-type group rather than an identified individual flat. Of the test rows, 99.2% have such history in the training data, with a median count of 15.

Both features are computed from the target, so they cause target leakage unless each value uses only rents that would be known at prediction time. A training row therefore receives the expanding mean over strictly earlier months, excluding its own month, so it uses neither its own rent nor the rent of a flat approved in the same month. A validation or test row receives the mean over all training rows, which is the information available at prediction time, when the most recent label is up to 16 months old. Rows without history keep a missing value, which the boosting model assigns to a learned branch, and the count indicates how reliable the mean is.

Block history lowers the RMSE by 12.1 on the rolling folds and by 13.7 on the confirmation cut, and it improves every fold. The improvement comes entirely from rows with history, since the two models perform equally on the 1 to 3% of fold rows without one. This model scored 436.05 on the public leaderboard, and Table 7 gives the results by fold.

Block history reduces mean under-prediction from S$80 to S$49. Earlier rents contain information about both the block premium and historical market levels. The reduction in bias is consistent with history partially compensating for the absence of explicit detrending, but does not isolate the source of the RMSE gain.

C. Target Encoding of Location

Target encoding replaces a category with a smoothed mean of the target [17]. We encoded detrended rent at seven location groupings involving town, street, block, subzone, and flat type. Group means are partially pooled toward broader means with a weight equivalent to twelve observations, reducing sensitivity to small groups. Training group aggregates exclude the row's approval month. Validation encodings use only outer-training rows. Two spatial rent priors average detrended rents from the ten and thirty nearest other blocks, excluding the block being encoded.

The configuration without hierarchical target encoding scores 523.32, compared with 529.30 when it is included. Changing the smoothing weight or using recency weights with 12- or 24-month half-lives changes RMSE by at most 1.8. All remain worse than the reference. A half-life is the interval after which an observation's weight is halved.

We tested whether the gap reflects the age of available history. Each training row received block history truncated a random 1–16 months before its approval month, approximating forecast-time staleness. RMSE increased by a further 12.5. This intervention did not improve accuracy and does not establish that staleness is irrelevant.

Three representations of block identity increase RMSE by 1.2–6.0 on the detrended target, while removing spatial priors costs only 0.1. These comparisons are consistent with substantial overlap between rental-history features and geographic

predictors. Small group sizes and within-group dispersion may also reduce the precision of block means. The final configuration retains the spatial priors and excludes hierarchical target encoding.

D. Macroeconomic Series

The project specification proposes COE premiums and share prices as economic indicators. We aggregated COE premiums by month and vehicle category and computed monthly share-price means, log changes, and sector baskets. Adding these features increases RMSE by 10.0 and 10.5 in two configurations. They vary by month rather than by flat, so their contribution concerns temporal conditions. These results do not establish useful forecasting information beyond the index.

We therefore tested whether they predict the market index, the only quantity that is extrapolated. In backtests from the three rolling cuts over the following 16 months, the forecast error is 1.26% of the index level for the damped trend, 1.87% for the property-sector basket, 1.90% for a random walk, and 2.61% to 5.14% for the COE series in levels. A regression on 12-month changes performs much worse, at 17.30%, because beyond the first year it uses its own forecasts as inputs and the errors accumulate. Every forecaster, including the macroeconomic regressions, is fitted on the index estimated from months before its cut, and only the realized index used for scoring uses all labeled months. The best macroeconomic forecaster, the property basket, improves on a random walk by only 0.03 percentage points, although COE premiums and the rent index correlate at 0.91 in levels. Both series rose steeply until 2023 and then flattened, and this shared trend produces the correlation.

E. Handling of Categorical Attributes

The six categorical columns are town (26 levels), flat type (six), flat model (20), planning area (31), region (five), and subzone (156). The eleven-model comparison retains these source attributes. LightGBM, CatBoost, and XGBoost use library-specific categorical handling. Ridge, nearest neighbors, and the neural network use one-hot indicators. The other tree models use integer category codes, whose arbitrary order constrains threshold splits. Table 5 specifies the full preprocessing. Encoding therefore remains part of each model pipeline rather than an independently controlled factor.

The category dictionaries of the training and prediction data are merged before fitting, so that a level absent from one side keeps its own code instead of receiving the code of another level. No level occurs only in the test file, but within a fold, where training ends up to 23 months earlier, this step affects the 156 sparsely populated subzones.

VII. Models and Ensemble

A. Model Families and Tuning

With the detrended target, we compared four model families on identical features and splits. CatBoost [11] scores 521.49, LightGBM [10] 523.32, XGBoost [12] 528.67, and one-hot ridge regression 534.26. The four configurations span approximately 13 RMSE. This comparison does not isolate model capacity.

We tuned the two leading libraries by random search with 50 draws, 28 for LightGBM and 22 for CatBoost, each evaluated under the protocol of Section III and run as one task of a cluster job array. The LightGBM draws range from 519.87 to 554.50 and the CatBoost draws from 520.19 to 523.87, so every CatBoost draw lies within 3.7 of the best. The selected LightGBM configuration uses 800 trees, a learning rate of 0.03, 127 leaves, 85% row sampling, 90% column sampling, and an L2 penalty of 5. Both selected configurations are stored in the repository. Tuning improved the RMSE by 3.45, and a later search found no configuration that improved the confirmation cut.

The length of the training history also affects accuracy. Restricting training to the most recent 18 or 24 months increases the RMSE by 20.2 and 3.7, while a 36-month history and the full history perform equally. After detrending, a 2021 rental remains informative about how one flat compares with another, so we train on all labeled months.

B. Averaging Ensemble

The final prediction is a weighted mean of three members. This is an averaging ensemble rather than stacking, since no meta-learner is trained on held-out predictions. Stacking would add another level of fold-internal fitting to folds that already re-estimate the index and the block history.

Non-negative weight search over the three boosting libraries gives 45% LightGBM and 55% CatBoost, with RMSE 518.34. Equal weights give 518.38. XGBoost receives no weight near the optimum. We use equal weights and average three LightGBM seeds and two CatBoost seeds to reduce sensitivity to initialization and sampling. Final predictions are clipped to the observed S$300–S$7,600 range. Across overlapping validation windows, clipping altered none of the 177,666 cached forecast instances from the repaired, single-seed validation blend. They range from S$1,278 to S$5,659, so its RMSE is unchanged.

The third member is HistGB trained on unadjusted rent with block history (Section VI-B). Its rolling RMSE is 527.0, compared with 519.9 for the LightGBM reference, but its predictions improve the average. At weight 0.30, the blend reduces rolling RMSE from 518.56 to 516.66 and confirmation RMSE from 466.54 to 463.77. With subsequent index and block-composition refinements, the corresponding scores are 515.73 and 462.54. These cached comparisons use one seed per boosting library and the historical inner-tuning protocol. The final refit averages seeds as described above. Appendix E reports the repaired protocol separately. The member trained on unadjusted rent provides a different temporal specification, although the contribution of that difference was not isolated.

Four further candidates were rejected because they improved the rolling folds but not the confirmation cut, namely a ridge member at 10%, the block-history member trained on our feature set, a weight of 0.25 instead of 0.30, and weights that vary with the horizon. In each case the folds improved by

a few tenths while the confirmation cut became worse, which is consistent with selection noise after many configurations have been compared on the same three windows.

C. Comparison of Eleven Model Classes

To broaden the comparison beyond the tuned boosting models, we evaluated eleven model classes without tuning. Each receives the same 29 source columns, comprising 23 numeric attributes and six categorical attributes represented as in Table 5. Block history, hierarchical target encodings, and block composition are excluded, and a global index is used. The classes comprise a town–flat-type median lookup, ridge regression, k-nearest neighbors with $k = 20$, a regression tree, bagged trees, random forest, AdaBoost, gradient boosting in three libraries, and a small neural network. In this setup, time enters through calendar month and approval-dependent age features. The explicit month counter is added only in the third treatment.

Each model is evaluated with index-adjusted rent, unadjusted rent, and unadjusted rent augmented by the explicit month counter. The unadjusted-rent treatment retains calendar month and time-dependent age features. This comparison tests whether the explicit temporal coordinate substitutes for index adjustment under otherwise fixed source attributes (Table 1).

Table 1. Each metric is averaged over three rolling folds for models trained on the same 29 source columns. Columns under With Index report index-adjusted results. The Unadj. column uses unadjusted rent, and + Month adds the month counter. Bias is the mean prediction error.

	With Index				Unadj.	+ Month
Model	**RMSE**	**MAE**	**Bias**	R^2	**RMSE**	**RMSE**
Median lookup	554.6	428.9	−38.9	0.36	965.0	n/a
Regression tree	540.6	412.6	−4.6	0.39	748.8	564.5
AdaBoost	535.9	415.1	−42.5	0.40	743.2	589.2
Ridge	534.4	410.5	−29.4	0.40	840.6	643.6
Neural network	530.8	407.8	−29.8	0.41	699.0	704.5
XGBoost	528.3	401.1	+1.7	0.42	603.7	567.2
Bagging	528.3	400.5	−1.4	0.42	658.1	555.8
Random forest	526.1	398.4	−2.1	0.42	647.2	549.7
k-nearest neighbors	523.8	397.4	−0.5	0.43	805.7	583.7
LightGBM	523.4	397.0	−0.5	0.43	611.8	556.2
CatBoost	522.3	397.8	−9.2	0.43	662.5	540.4

Under index adjustment, the best five models differ by 5.9 RMSE. Untuned k-nearest neighbors is 1.5 above CatBoost and the median lookup is 32.2 above it. The largest R^2 is 0.43. Mean absolute error (MAE) gives a different order. LightGBM and k-nearest neighbors precede CatBoost. The lookup, ridge, and neural network under-predict by S$29–S$39 on average. Bagging, random forest, LightGBM, and XGBoost have mean errors within S$3. Bagging scores 12.4 below the single tree and 2.2 above the random forest, but different leaf settings prevent attribution of the gain to averaging alone. AdaBoost scores 535.9, compared with 534.4 for ridge.

On unadjusted rent, the range across models is 361.4 RMSE and every model under-predicts on average. Gradient boosting performs best among these configurations. Although the explicit month counter is excluded, lease age and block age retain temporal information, so this treatment does not remove all date-related predictors.

Adding the month counter does not match index-adjusted accuracy for any evaluated model. Ridge and the neural network over-predict by approximately S$290 and S$400. Tree-based models and nearest neighbors under-predict. These results describe the evaluated specifications and do not establish that other direct temporal models would perform similarly.

VIII. Error Analysis

A. Development Results

Figure 5 summarizes the historical development record. The retained sequence reduces rolling RMSE from 539.7 to 515.7 over seven stages. It combines changes to features, index specification, and ensemble construction, so the 24-point gain is a pipeline comparison rather than a sum of independent effects. Rejected alternatives are evaluated against their contemporaneous references. Training-window restrictions and lagged encodings also produce substantial changes. Feature and model refinements generally have smaller effects than omitting index adjustment.

B. Within-Group Variation and Index Error

We analyse the selected LightGBM configuration using pooled within-group variation and an oracle-index comparison. These are diagnostic estimates rather than independent causal components of prediction error.

Within groups containing at least two validation rows with identical predictors, we estimate noise variance by dividing the within-group sum of squares by the number of observations minus the number of groups (Appendix H). This correction accounts for estimation of each group mean. Interpretation as conditional noise requires independent within-group errors around a common mean and representative group variances. An oracle variant estimates the index using both training and validation rents before refitting the model. The Noise column reports estimated within-group variance as a percentage of model mean squared error (MSE). The Oracle column reports the reduction in MSE obtained with the oracle index, as a percentage of model MSE. Resid. denotes the remaining share and equals 100% minus Noise minus Oracle. All three shares use only repeated-group rows. They sum to 100% before rounding but do not identify independent causal sources of error.

Across the three rolling folds, the estimated noise, oracle-improvement, and residual shares are 98.8%, 1.1%, and 0.1%, respectively (Table 2). The pooled noise standard deviation is S$509, compared with model RMSE S$512 on the same rows. Negative residual shares in some folds reflect estimation uncertainty rather than negative error variance. These rows account for 27.6% of validation observations. The estimates do not establish an error floor for other rows or the hidden test set.

The closeness of estimated noise and model error is consistent with limited gains among the tested models on these groups. It does not exclude improvements from different features, learners, or temporal assumptions. Similar aggregate

Table 2. Error decomposition is reported for the selected LightGBM configuration. Under All Rows, Oracle is RMSE using an index fitted on training and validation rents. Under Rows in Repeated Groups, Noise SD is the estimated within-group standard deviation, and Noise, Oracle, and Resid. are the MSE shares defined in Section VIII. RMSE and Noise SD are in Singapore dollars. The summary row averages fold RMSEs for All Rows and pools forecast instances for repeated groups.

	All Rows		Rows in Repeated Groups				
Fold	**RMSE**	**Oracle**	**RMSE**	**Noise SD**	**Noise**	**Oracle**	**Resid.**
1	554.7	553.1	544.9	542.5	99.1%	0.3%	0.5%
2	519.2	514.9	511.5	506.1	97.9%	2.2%	0.0%
3	482.6	481.1	476.3	475.4	99.6%	0.8%	−0.4%
Summary	519.9	517.5	512.0	509.0	98.8%	1.1%	0.1%
Confirm	465.5	463.1	456.5	457.4	100.4%	0.4%	−0.8%

errors on grouped and ungrouped rows do not establish equal conditional noise in the two populations.

The oracle reduces all-row RMSE by 1.5–4.2 across the rolling folds, compared with the 86.4-point cost of omitting index adjustment. This suggests that rescaling historical rents provides most of the measured benefit under these windows. Because the oracle uses validation targets and changes the transformed model, its improvement is an optimistic diagnostic rather than a formal bound on gains from forecasting.

Estimated within-group noise standard deviations are S$279, 471, 588, 479, and 443 for 2021 through early 2025. Conditional dispersion therefore varies over time, which may contribute to differences between validation periods. This does not establish stability of the conditional mean or explain the lower public score, since test-period noise is unobserved.

C. Error by Flat Type and Limitations

RMSE increases from S$483 for 3-room to S$624 for executive flats. Relative RMSE decreases from 22% of mean rent for 2-room to 16% for 5-room flats. Mean errors for these types range from S$34 over-prediction for 2-room flats to S$20 under-prediction for executive flats. The 1-room group has only 46 fold observations and mean error S$115 (Table 8). Larger conditional errors increase the contribution of expensive rentals to squared-error evaluation.

The analysis has six limitations. Noise is estimated only for repeated feature groups, which may not represent other rentals. The oracle uses validation targets and is not an attainable forecast. All rolling origins follow the 2022 surge, limiting evidence for a renewed rising market. Unobserved flat attributes can explain variation classified here as noise. The confirmation cut influenced selection, and repeated comparisons on overlapping windows create selection uncertainty. Finally, the historical block-history member used row-based inner splits and features constructed before those splits. The repaired month-based protocol rebuilds history within each inner fold. It changes member rolling RMSE from 527.0 to 526.4 and blend RMSE from 516.66 to 516.54 (Appendix E).

D. Implications for Stakeholders

For tenants, the model provides comparisons conditional on recorded flat attributes. The estimated S$509 noise standard deviation describes variation within repeated feature groups, not a calibrated prediction interval for an individual listing. Condition and tenancy terms remain unobserved.

For landlords and agents, town and flat type have greater held-out permutation importance than nearest-MRT distance. These values measure predictive reliance rather than causal effects or returns to advertising decisions. The estimated market indices also distinguish temporal movements across flat types, rising by 46% for 2-room and 61% for executive flats over the labeled period.

For policy makers, the index adjusts estimated market movements for observed block and flat-type composition under the stable-base assumption. Within-group dispersion rises from S$279 in 2021 to S$588 in 2023 before declining to S$443 in early 2025. This describes changes in conditional dispersion without identifying their causes or the effects of a cooling measure.

E. Future Work

Future work should compare the additive and multiplicative decompositions under separate hyperparameter searches with equal validation budgets. Additional chronological origins should cover periods of both rising and stable rents to test the damping assumption, with a separate later window reserved for evaluation after model selection. Uncertainty in model differences should be assessed by paired resampling of contiguous calendar-month blocks, retaining all forecasts of an approval together across overlapping validation windows. Geographic features should incorporate station opening dates so that nearest-station distances reflect the operating network in each approval month. Floor level, flat condition, tenancy duration, and included services should also be collected where available at prediction time. These attributes could distinguish rentals that currently share identical predictors. Re-estimating within-group variation after adding them would test whether part of the estimated noise reflects omitted attributes. Prediction intervals should also be evaluated for coverage by forecast horizon and flat type before use in individual rental decisions.

IX. Conclusion

The temporal test split motivates separate modeling of market movements and differences between flats. In the baseline holdout, the month counter has zero permutation importance because its future values follow the same tree branches. An explicit market-index forecast addresses this limitation of the time feature while retaining tree models for the remaining regression.

In matched LightGBM comparisons, index adjustment reduces rolling RMSE by 86.4, whereas tuning improves it by 3.5. Eleven model classes span 32.2 under a fixed index setup, with the best five within 5.9. On the 27.6% of validation rows in repeated feature groups, estimated pooled noise variance equals 98.8% of one LightGBM configuration's MSE. This result is conditional on the grouping and estimation assumptions and does not establish a test-set error floor. The ensemble scores S$431.01 at shrinkage 0.5, and its 0.7 variant achieves our best public RMSE of S$430.76.

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APPENDIX

A. Workload Breakdown

Table 3 lists the contribution of each member.

Table 3. The workload is summarized by team member and primary responsibility.

Member	Main Responsibilities
Angeles Mathena	Modeling pipeline, market index and its forecast, ablation study, error decomposition, and final ensemble
Alzamil Nojood Ahmad	Feature extensions, macroeconomic series, and index variants
Su Yubo	Data integration, geographic join repair, block history features, validation design, and the model comparison
Szentpeteri David	Exploratory analysis, data cleaning, first feature set, and baseline models
Dai Peisong	Exploratory analysis, feature construction, model comparison, and ablation study

B. Validation Splits

Table 4. The first three rows describe the rolling validation windows, and the last row describes the confirmation window. Lengths are given in months.

Training Months	Validation Months	Length	Rows
Jan 2021 to Mar 2023	Apr 2023 to Jul 2024	16	48,301
Jan 2021 to Jul 2023	Aug 2023 to Nov 2024	16	47,036
Jan 2021 to Nov 2023	Dec 2023 to Mar 2025	16	47,171
Jan 2021 to Mar 2024	Apr 2024 to Mar 2025	12	35,158

C. Representation of the Categorical Attributes

Table 5. The six categorical columns are represented separately for each model family in the eleven-model comparison.

Model Family	Representation
LightGBM, CatBoost, XGBoost	Library-specific categorical handling, without treating category labels as numeric magnitudes.
Regression tree, random forest, bagging, AdaBoost	Ordinal integer codes. Unseen levels map to -1. The arbitrary ordering constrains threshold splits and may require extra splits to represent category-specific effects.
Ridge, nearest neighbors, neural network	One-hot indicators, levels appearing fewer than 30 times pooled into one column, with median imputation and standardization of the numeric columns. These three would read an integer code as a magnitude.

D. Permutation Importance of the Gradient-Boosting Baseline

Table 6. Importance is the decrease in held-out R^2 after one feature is shuffled in the chronological baseline holdout.

Feature	Importance
Town	0.352
Flat type	0.261
Floor area	0.134
Lease age	0.061
Flat model	0.021
Distance to nearest MRT station	0.012
Distance to nearest mall	0.007
Distance to nearest primary school	0.001
Calendar month	0.0003
Month counter	0 (exact)
Mean stock closing price	-0.00001
Mean COE premium	-0.00003

E. Block Rent History by Fold

Table 7. RMSE and mean prediction error are reported for gradient boosting trained on unadjusted rent with and without block history. The final row reports the confirmation cut.

Validation Months	RMSE		Mean Error	
	Without	With	Without	With
Apr 2023 to Jul 2024	585.4	572.8	−139.8	−120.8
Aug 2023 to Nov 2024	532.5	521.7	−36.1	−8.7
Dec 2023 to Mar 2025	499.5	486.6	−64.6	−16.4
Mean of the three	539.1	527.0	−80.1	−48.7
Apr 2024 to Mar 2025	481.5	467.8	−56.5	−28.8

The **With** column and the submitted blend use positional `TimeSeriesSplit` for inner tuning. Its boundaries can divide a month, evaluating within-month interpolation as well as future prediction. Block-history features were also constructed before inner splitting and could include inner-validation rents. The repaired protocol uses whole-month origins and rebuilds history from each inner training frame. Its horizon is 16 months where feasible and 14 at the earliest cut. Both protocols use identical sampled candidates. The comparison measures the combined repair of inner splits and history construction.

The repair changes the selected setting at two of the four origins, cuts 27 and 39, where the positional search had selected a learning rate of 0.03 rather than 0.1. At the other two origins it selects the same setting and the fold predictions are identical. Where the selection changes, the member improves by 1.87 at cut 27 and 0.22 at cut 39, so its rolling mean moves from 527.02 to 526.40 and the confirmation cut from 467.76 to 467.54. The ensemble moves from 516.66 to 516.54 on the folds and from 463.77 to 463.82 on the confirmation cut, and 30% remains the best weight.

In this comparison, the repair improves the member's outer-validation scores where selection changes. The ensemble changes by less than 0.2 RMSE on either protocol. This sensitivity analysis does not establish that inner leakage generally has negligible effects. `src/h2h.py --inner rows` reproduces the historical column. `--inner month` writes the repaired results to `outputs/ablation_head_to_head_repaired.csv`.

F. Error by Flat Type

Table 8. Errors for the selected LightGBM configuration are pooled over the rolling folds and reported by flat type. Overlapping validation windows can count an approval more than once.

Flat Type	Rows	Mean Rent	RMSE	Bias	RMSE / Rent
1-room	46	1,596	439.8	+114.6	27.6%
2-room	3,637	2,184	485.0	+34.3	22.2%
3-room	46,708	2,696	483.3	+18.9	17.9%
4-room	51,844	3,237	527.4	−12.9	16.3%
5-room	32,901	3,440	536.3	−17.3	15.6%
Executive	7,372	3,577	623.7	−19.6	17.4%

G. Mean Test-Period Index Under Each Forecast

Table 9 gives the values behind the 1.5% range in Section V-B, computed by `src/index_horizon.py`. The mean index is taken over the 50,000 test rows, so it is weighted by the number of rows at each horizon. The shift is the difference between each mean index and that of the submitted forecast, multiplied by the mean training rent of S$2,719.

Table 9. Each mean index is weighted by the number of test rows at each forecast horizon. The reported shift is the difference from the submitted forecast's mean index, multiplied by the mean training rent of S$2,719.

Forecast	Mean Index	Shift (S$)
Random walk	1.0000	−6.6
Damped, 12 months, ϕ = 0.4	1.0011	−3.5
Damped, 12 months, ϕ = 0.6 (shipped)	1.0024	0.0
Damped, 12 months, ϕ = 0.75	1.0043	+5.2
Damped, 12 months, ϕ = 0.85	1.0068	+11.8
Damped, 12 months, ϕ = 0.95	1.0113	+24.0
Damped, 18 months, ϕ = 0.85	1.0063	+10.6
Linear, 12 months	1.0150	+34.2
Linear, 18 months	1.0141	+31.7

H. Pooled Variance Estimator

For groups g containing $n_g \geq 2$ rows with identical predictors, the pooled variance estimator is

$$\widehat{\sigma}^2 = \frac{\sum_g \sum_{i \in g} (y_i - \bar{y}_g)^2}{\sum_g (n_g - 1)}.$$

The denominator removes one degree of freedom per fitted group mean. Groups are defined separately within each validation fold to avoid counting overlapping observations as additional within-group matches. This estimator describes pooled within-group variation. Interpreting it as the noise level for other observations requires additional representativeness assumptions.

Multiplying the training variance estimate by the proportion of rows in repeated groups and taking the square root gives approximately S$240. Applying it to the test-group proportion gives approximately S$245. These illustrative scales assume zero error on all remaining rows. The test calculation additionally transfers historical noise into an unobserved period. Neither calculation is an empirical test-set RMSE or a rigorous lower bound.

I. The Distribution of Rent

Figure 3 uses all labeled rentals to match the training sample analysed in Section II-B.

J. Detailed Ablation Results

Figure 5 summarizes the ablation record of Section VIII-A, showing the retained stages and the cost of the rejected alternatives. Figure 4 gives the results for individual candidates. Panel (a) is the feature-group ablation, panel (b) gives the gradient-boosting configurations by fold, and panel (c) compares the rolling-fold score of each candidate with its confirmation-cut score.

K. Reproducing Each Result

The repository contains scripts, saved results, and reproduction instructions in `README.md`. First, `src/data.py` builds `outputs/base.parquet`. Analyses then read this feature

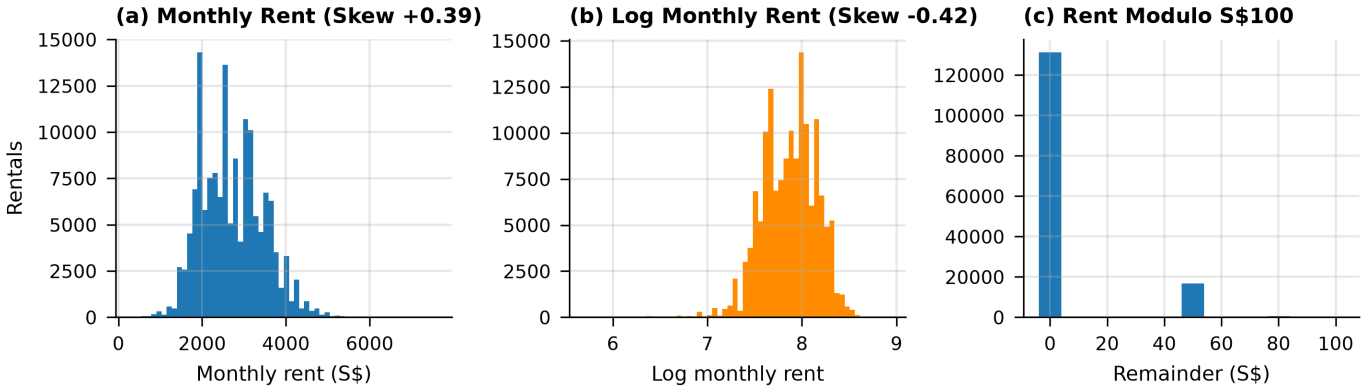


Figure 3. All panels use the 150,000 labeled rentals. Panel (a) shows the mildly right-skewed rent distribution. Panel (b) shows the left-skewed distribution after logarithmic transformation. Panel (c) shows that rents follow a coarse grid, with 98.7% being exact multiples of S$50.

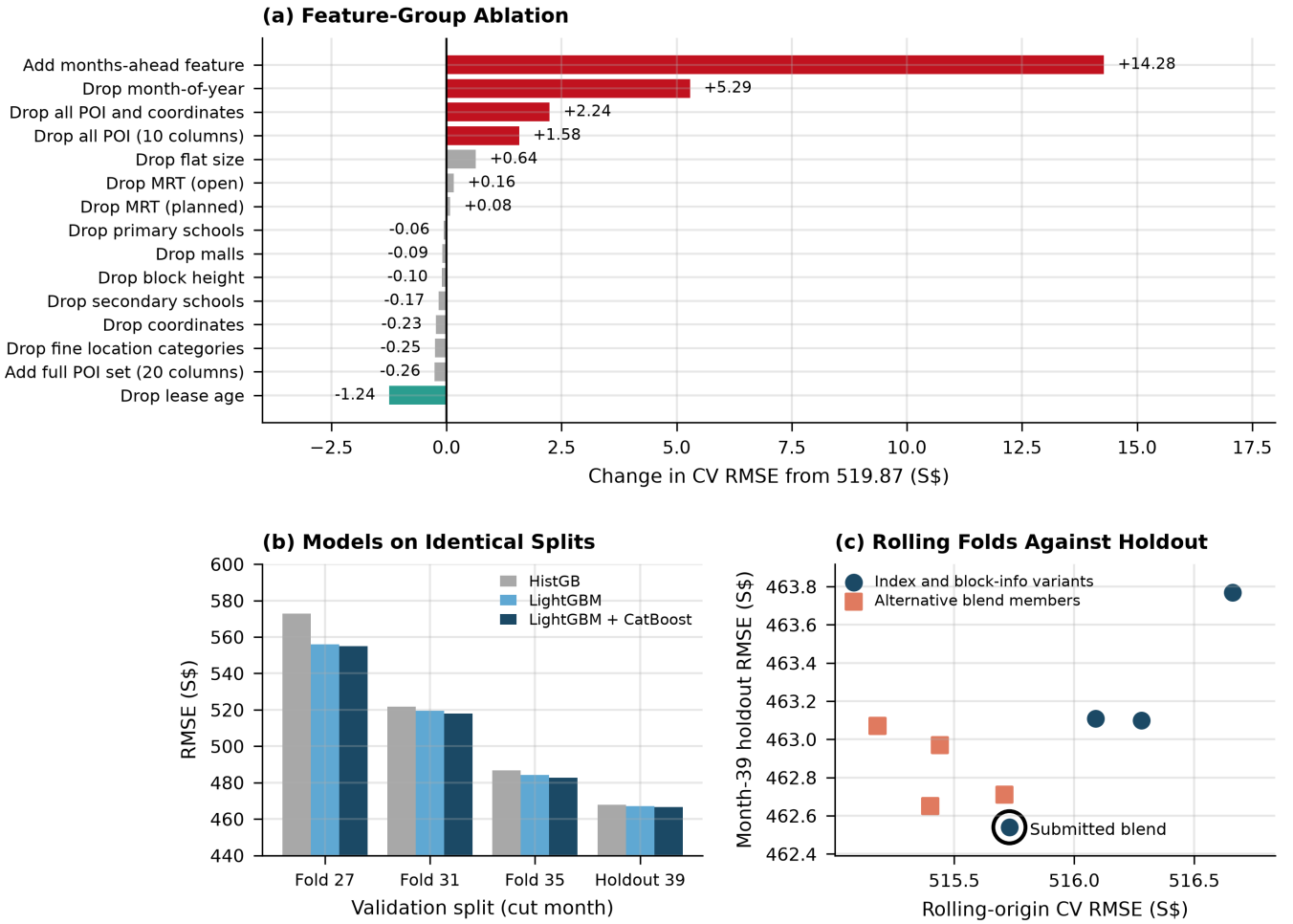


Figure 4. Panel (a) shows the RMSE change after adding or removing each feature group relative to the tuned LightGBM reference of 519.87. Panel (b) compares three configurations across validation splits, including the March 2024 confirmation cut. Panel (c) compares candidates' rolling-fold and confirmation-cut RMSEs.

table. Ensemble analyses have additional dependencies. Scripts `h2h.py`, `confirm.py`, and `members.py` generate caches used by the blend scripts. Figure scripts run after their input CSV files exist. Table 10 maps results to scripts and output files. Historical replay limitations are specified below.

For historical ablation tables, `src/groups.py` maps each named comparison to registry configurations. `run_ablation.py --group` *name* re-evaluates the group un-

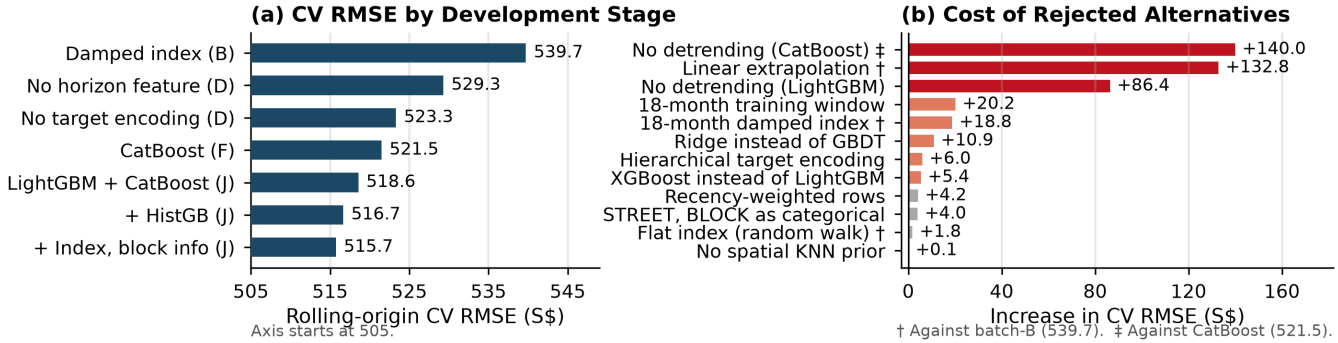


Figure 5. Panel (a) shows rolling-origin RMSE after each retained development stage, labeled by ablation batch. Panel (b) shows the RMSE increases for rejected alternatives. The reference is 523.3 except for batch B at 539.7 (†) and CatBoost at 521.5 (‡).

Table 10. Each reported result is mapped to its generating script and saved output.

Where	Script in `src/`	Output in `outputs/`
Sec. II, App. I	`eda.py`	`eda_summary`
Sec. IV, App. D	`baseline.py`	`permutation_importance`
Sec. V-B, Figure 2	`run_ablation.py`	`ablation_index_forecast, ablation_index_window_damping`
Sec. V	`additive_check.py`	`additive_index_check`
Sec. V-B, App. G	`index_horizon.py`	`index_test_horizon`
Sec. V-C	`run_ablation.py, index_horizon.py`	`ablation_index_segments, index_by_flat_type`
Sec. VI	`run_ablation.py, altreps.py`	`ablation_feature_groups, ablation_location_representation, ablation_altreps`
Sec. VI-B, App. E	`blockhist.py, h2h.py`	`ablation_blockhist, ablation_head_to_head`
Sec. VI-C	`run_ablation.py`	`ablation_target_encoding, ablation_lagged_block_history`
Sec. VI-D	`macro.py`	`macro_backtest`
Sec. VII	`run_ablation.py, tune.py`	`ablation_model_family, tuning, ablation_retune`
Sec. VII-B	`blend_weights.py, blend_members.py`	`ablation_blend_weights, ablation_blend_members`
Sec. VII-C, Table 1	`ladder.py`	`ladder_metrics, ladder_by_horizon`
Sec. VIII, Table 2	`confirm.py, error_decomp.py`	`ablation_confirmation_cut, error_budget`
Sec. VIII, Apps. F, H	`residuals.py`	`residual_slices, residual_by_horizon, rmse_floor, feature_floor`
Sec. V	`compare_submissions.py`	`submission_comparison`
Final predictions	`predict.py`	`submissions/`

der current defaults and writes a separate `_rerun` file. Batch B preserves the ranking but changes the scores. Its flat-index configuration gives 526.6 instead of 541.5. Batch D reproduces all nine rows at their recorded precision, including fold scores and bias, after explicitly specifying the historical encoding mode. `--list-groups` reports coverage. Seven groups contain registered configurations only, while four also contain historical variants absent from the registry. Encoder settings must be explicit because smoothing and recency parameters are inactive when target encoding is disabled. The original tables retain the comparisons used during development.

Forty of the fifty historical hyper-parameter draws regenerate from the current sampler and seed. Ten draws, including the selected CatBoost configuration, predate that sampler. Frozen parameters in `src/tuned_params.py` support retraining the reported models.